Two circular garden beds each have radius 5 m. Their centres O_1, O_2 are 6 m apart. The circles intersect at A, B. The segments AB and O_1O_2 are perpendicular and bisect each other at M. Let $\alpha = \angle AO_1M$. Figure 5 shows the common area of the two beds.

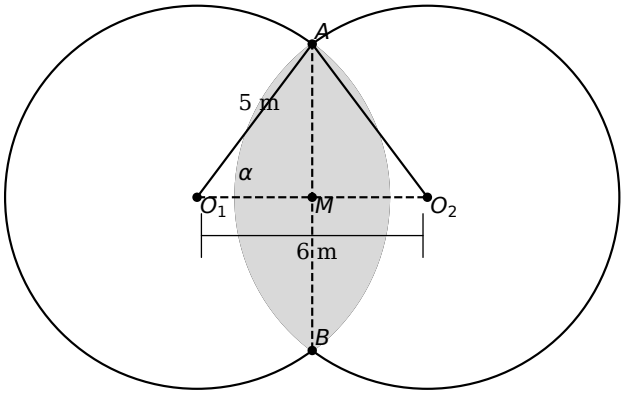


Figure 5

(a) Find AB and give an exact expression for α. (3)

(b) Find the common area, giving your answer in terms of α. (3)

(c) Find the perimeter of the common area, to 3 significant figures. (2)

(Total for Question 5 is 8 marks)

A road of width 4 m consists of two annular sectors joined end to end along a complete radial edge. Each sector has inner radius 8 m, outer radius 12 m and angle $\pi/3$. The first bends left and the second bends right, so that each long side contains one inner arc and one outer arc. Figure 15 shows the plan.

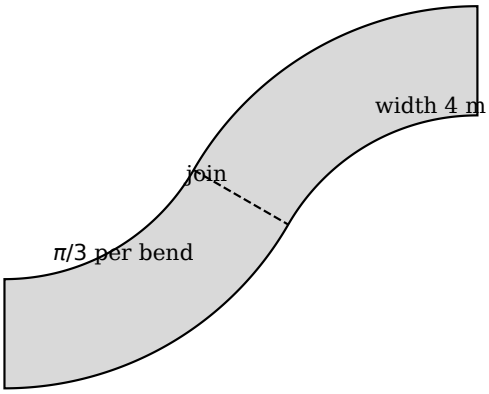


Figure 15

(a) Find the length of each curved long side exactly. (2)

(b) Find the total road area exactly. (2)

(c) Edging is required around the whole outside boundary, including the two open ends. Find the length required. (3)

(Total for Question 15 is 7 marks)

A sector-shaped garden of radius r m is divided into two smaller sectors by one radial fence. Their angles are θ and 2θ. All the outer boundary and this internal fence use 54 m of fencing. The garden has area $108\ \text{m}^2$ and total angle less than π.

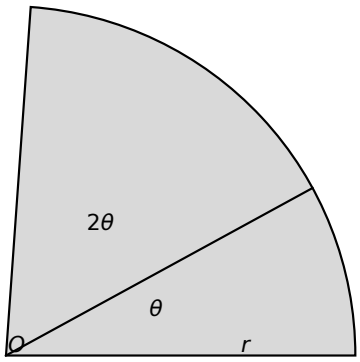


Figure 24

(a) Show that $r^2 - 18r + 72 = 0$. (3)

(b) Find the radius and both smaller angles, explaining which root must be rejected. (3)

(c) Find how much fencing is around the outside boundary only. (2)

(Total for Question 24 is 8 marks)

Figure 5 shows $y = \sqrt{x}$, the line $y = x - 2$, and the x-axis. The shaded finite region is bounded by all three.

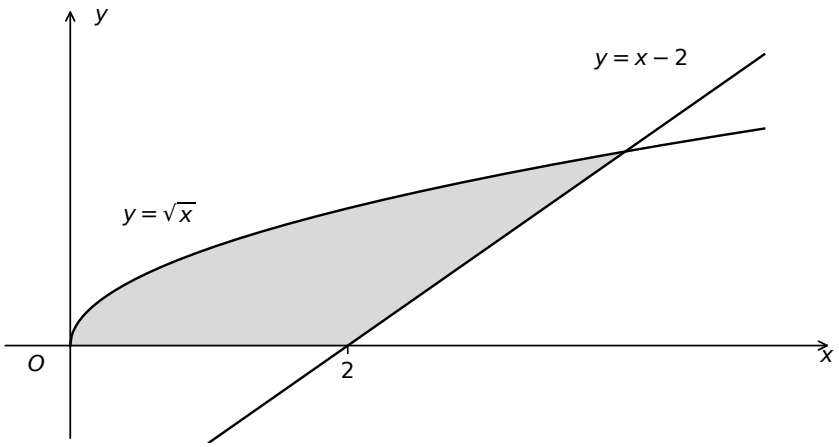


Figure 5

(a) Find the intersection of the curve and line, checking the sign condition before squaring. (3)

(b) Find the exact shaded area. (3)

(c) Explain why $\int_0^4 [\sqrt{x} - (x - 2)]dx$ gives too large a result. (2)

(Total for Question 5 is 8 marks)

A channel cross-section has depths, in metres, measured below a horizontal water surface:

x **(m)**	0	2	4	6	8
depth	0	0.8	1.6	1.4	0

For this question, model the bed by straight line segments joining consecutive measurement points. The water surface is then lowered by 0.2 m.

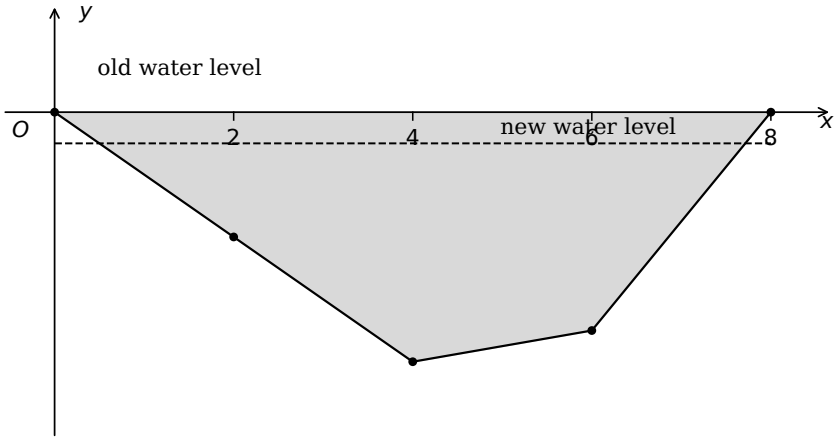


Figure 10

(a) Find the original cross-sectional water area. (2)

(b) Find the new positions of the left and right water edges. (3)

(c) Find the new cross-sectional area. Explain why simply subtracting $8(0.2)$ from the old area is not exact. (4)

(Total for Question 10 is 9 marks)

A planting area is bounded by $y = 4$, $y = 1 + \sqrt{x}$ and the y-axis, as shown in Figure 24. Coordinates are in metres. A project is viable only if the planting area is at least $9.5\ \text{m}^2$.

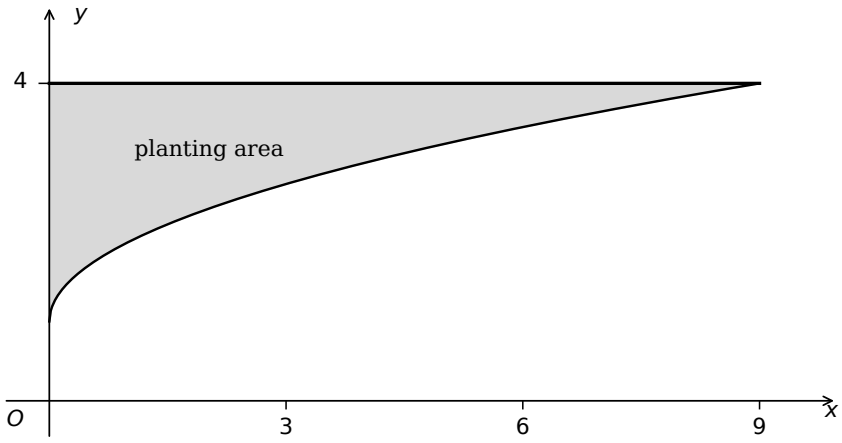


Figure 24

(a) Using the points $x = 0, 3, 6, 9$ on the curved boundary, obtain a trapezium-based estimate of the planting area. (4)

(b) Use integration to find the exact planting area. (3)

(c) State the correct project decision and explain why using the estimate alone could lead to the wrong decision. (2)

(Total for Question 24 is 9 marks)

The function $f(x) = 4 - (x - 1)^2$ has domain $[-1, 4]$. Define $g(x) = |f(x)|$ on the same domain, and $h(x) = f(|x|)$ wherever this is defined. Figure 2 shows the original graph.

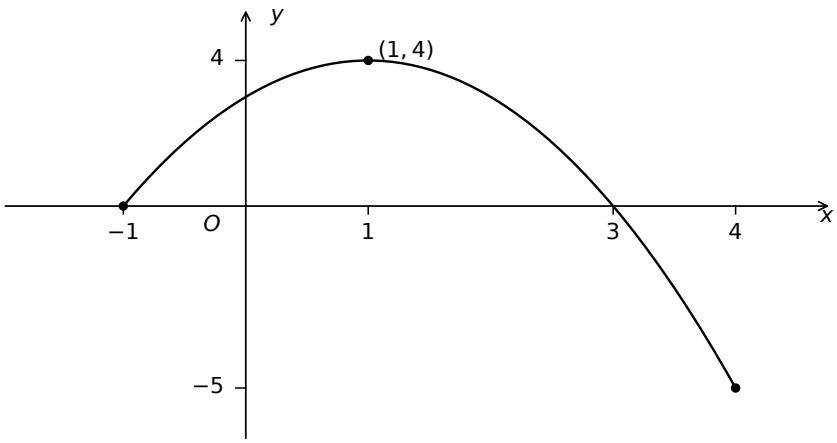


Figure 2

(a) Find the domain and range of h and sketch its graph, labelling its zeros and maximum points. (4)

(b) Find the range of g. (2)

(c) Find all k for which $y = g(x)$ and $y = k$ have exactly two distinct intersections. (3)

(Total for Question 2 is 9 marks)

The function is defined by $f(x) = x + 2$ for $-2 \leq x \leq 1$, and $f(x) = 5 - x$ for $1 < x \leq 4$. Figure 10 shows the open and closed endpoints.

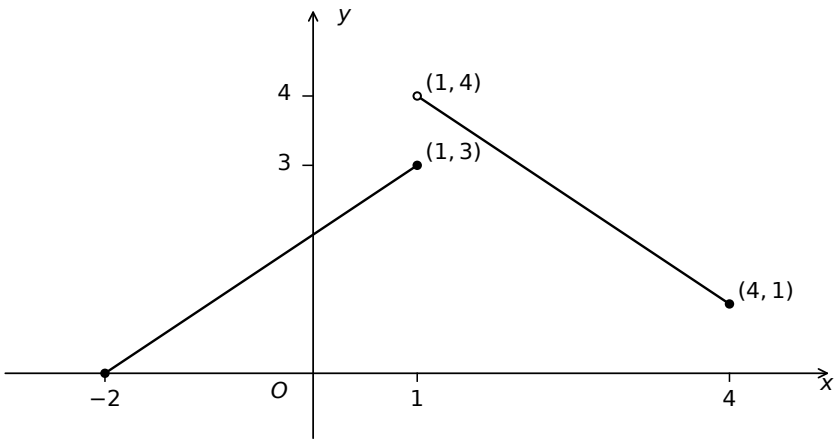


Figure 10

(a) Find the range of f and $ff(1)$. (3)

(b) Does f attain a greatest value? Explain using the endpoint at $x = 1$. (2)

(c) Restrict the domain to $[-2, 3/2]$. Find the inverse function in piecewise form, with its complete domain. (4)

(Total for Question 10 is 9 marks)

The complete graph of f is the two straight segments joining $(-2, -1)$ to $(0, 2)$ to $(2, -1)$, as shown in Figure 24.

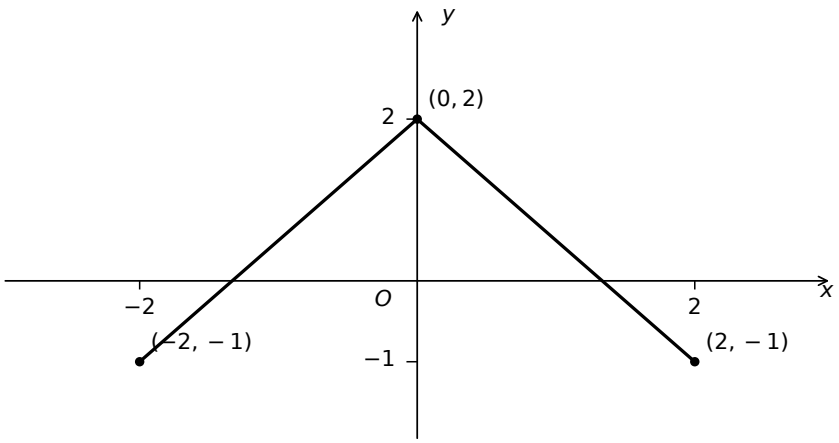


Figure 24

(a) Write an expression for f and state its range. (3)

(b) Restrict f to $[0, 2]$ and find the inverse of this restriction, including its domain. (3)

(c) Using the original full-domain f, find every solution of $ff(x) = 0$. (4)

(Total for Question 24 is 10 marks)

Question 5

P1-C07-R05 · 8 marks

(a) · 3 marks

$AB = 8$ m, $\alpha = \arccos(3/5)$.

公共弦在两圆心中点处，三角形O_1MA直角，$O_1M=3$、斜边5，得$AM=4$。弦长$2AM=8$，$\cos\alpha=3/5$。

M1	二等分对称图形及勾股
A1	弦长
A1	弧度角表达式

(b) · 3 marks

$50\alpha - 24$ m^2.

重叠区由两个相同弓形组成；每个扇形角2α，面积25α，减去底8、高3的三角形面积12，再乘2。

M1	识别两个弓形
M1	扇形与三角形的面积
A1	合并正确

(c) · 2 marks

$20\alpha \approx 18.5$ m.

边界是两条角2α、半径5的圆弧；公共弦AB在内部，不应计入。

M1	只计两圆弧
A1	周长及精度

Question 15

P1-C07-R15 · 7 marks

(a) · 2 marks

$20\pi/3$ m.

每条长边各含8×π/3和12×π/3，不是把两外弧归到同一边。

M1	每侧配一内弧一外弧
A1	两边共同长度

(b) · 2 marks

$80\pi/3$ m^2.

一段环形扇形面积½(144−64)π/3=40π/3，共两段，不重叠。

M1	一段环形扇形面积
A1	合计面积

(c) · 3 marks

$40\pi/3 + 8$ m.

两条长边共40π/3，只有首末两条宽4的径向边属于外界；中间接缝不计入。

M1	两曲边合计
M1	区分两端与内部接缝
A1	完整边界长度

Question 24

P1-C07-R24 · 8 marks

(a) · 3 marks

From $3r + 3r\theta = 54$ and $\frac{1}{2}r^2(3\theta) = 108$, eliminate θ to obtain the equation.

围栏含三条径向边，总长3r，曲边总长3rθ。由面积得3rθ=216/r，代总长式乘r并除3。

M1	三条径向围栏及曲边关系
M1	面积关系并消元
A1	推出二次式

(b) · 3 marks

$r = 12$ m; $\theta = 0.5$, $2\theta = 1$ radian. Reject $r = 6$, which gives total angle $6 > \pi$.

二次式分解为(r−6)(r−12)。总角216/r^2；r=6给6弧度，违背总角小于π；r=12给1.5，故分角0.5、1。

M1	两个候选半径
M1	检查总角限制
A1	正确半径和两角

(c) · 2 marks

42 m.

从总围栏54减一条内部半径12，得42；也可算曲边18加两外半径24。

M1	排除内部围栏
A1	外周长度

Question 5

P2-C08-R05 · 8 marks

(a) · 3 marks

$(4, 2)$.

√x=x−2要求x≥2。平方给x²−5x+4=0，候选1和4；1使原式右边为负，故舍弃。

M1	写出平方前的符号限制
M1	解二次式
A1	只保留有效交点

(b) · 3 marks

$10/3$ square units.

曲线下0到4面积16/3，减去直线与横轴之间x=2到4的三角形面积2，得10/3。若分段积分，0至2下边界是横轴。

M1	正确分界x=2或减去三角形
M1	正确积分曲线
A1	面积

(c) · 2 marks

For $0 < x < 2$ the line is below the x-axis; that integral includes an extra triangle of area 2.

所求阴影在0至2以下边界为横轴，不是线y=x−2；错误积分额外包含横轴以下底2高2的三角形。

B1	识别错误区间和下边界
B1	额外面积2

Question 10

P2-C08-R10 · 9 marks

(a) · 2 marks

$7.6\ \mathrm{m}^2$.

分段直线模型下梯形求和给准确模型面积：2/2[0+0+2(0.8+1.6+1.4)]=7.6。

M1	正确步长与梯形权重
A1	原面积

(b) · 3 marks

$x = 1/2$ m and $x = 54/7$ m.

左岸首段深度=0.4x，达到0.2处x=0.5；右岸最后段深度=0.7(8−x)，达到0.2处x=8−2/7=54/7。

M1	用左岸线性深度关系
M1	用右岸线性深度关系
A1	两个新边界

(c) · 4 marks

$851/140 \approx 6.08\ \mathrm{m}^2$.

初算7.6−1.6=6，把已露出的两岸也当成负水深；需加回两小三角形，面积½×0.5×0.2=0.05和½×(2/7)×0.2=1/35。总6+1/20+1/35=851/140。

M1	从旧面积减矩形或重新分段求和
M1	正确处理新干燥的两岸
A1	新面积
B1	说明固定8m宽扣法的错误

Question 24

P2-C08-R24 · 9 marks

(a) · 4 marks

$45/2 - 3\sqrt{3} - 3\sqrt{6} \approx 9.96\ \text{m}^2$.

先估曲线下方面积，步长3，估计3/2[1+4+2(1+√3+1+√6)]；从宽9、高4的矩形36中扣除。

M1	正确节点高度与步长
M1	梯形下方面积
M1	矩形补集
A1	估计面积

(b) · 3 marks

$9\ \text{m}^2$.

曲线与y=4在x=9相交，面积 $\int_0^9 (3 - \sqrt{x})dx = 27 - 18 = 9$。

M1	正确差函数及界
M1	正确积分
A1	精确面积

(c) · 2 marks

Not viable: $9 < 9.5$, although the estimate is above 9.5.

估计约9.96会误判达标；它是上方补集的高估，准确积分9低于规定门槛。

B1	正确决定
B1	估计可能误导的数值依据

Question 2

P3-C02-R02 · 9 marks

(a) · 4 marks

Domain $[-4, 4]$, range $[-5, 4]$; zeros $x = \pm 3$; maxima $(\pm 1, 4)$; $h(0) = 3$.

|x|只取原定义域中的[0,4]，再将这一半关于纵轴反射。不能把原图x<0的部分也保留。右端值−5仍是最小值。

M1	选取原图非负横坐标部分并反射
A1	定义域和值域
A1	两零点
A1	最大点及拼接点

(b) · 2 marks

$[0, 5]$.

原函数最小−5、最大4且过轴；负值绝对值后能到5，故新最大值不是4。

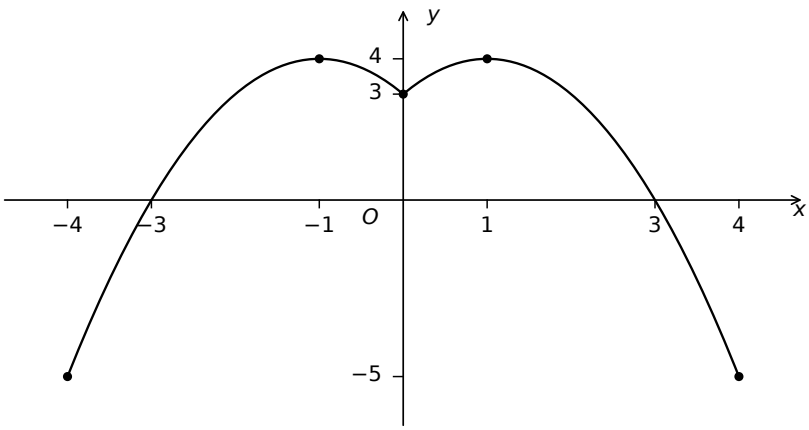
M1	比较原正最大值与负最小值绝对值
A1	完整值域

(c) · 3 marks

$k = 0$ or $k = 4$.

k=0时原零点−1、3两点；0

M1	按图分高度区间并处理极值
A1	k=0
A1	k=4且无额外值



Suggested sketch

Question 10

P3-C02-R10 · 9 marks

(a) · 3 marks

Range $[0,4)$; $ff(1)=2$.

第一段取[0,3]；第二段取[1,4)，合并[0,4)。f(1)=3采用第一段，再算f(3)=2。

M1	正确处理两段端点
A1	完整半开值域
A1	复合值

(b) · 2 marks

No. Values approach 4 from below as $x \to 1^+$, but 4 is excluded and $f(1)=3$.

靠近1右侧可输出比任何给定小于4的候选值更大的数，但4本身没有达到；不能把空心点当作函数值。

B1	未取得最大值
B1	空心端点及f(1)的解释

(c) · 4 marks

$f^{-1}(x)=x-2$ for $0 \le x \le 3$; $f^{-1}(x)=5-x$ for $7/2 \le x < 4$.

第二段现在输入(1,1.5]、输出[3.5,4)，与第一段输出[0,3]不重叠，因此限制后是一一映射。两支分别反解并保留输出域中的缺口。

M1	检查两输出段不重叠
M1	分别反解
A1	两支公式
A1	两个区间及缺口

Question 24

P3-C02-R24 · 10 marks

(a) · 3 marks

$f(x) = 2 - 3|x|/2$ on $[-2, 2]$; range $[-1, 2]$.

顶点在(0,2)，左右斜率大小均3/2且向下，故表达式如此；图像连续覆盖−1到2。

M1	由图恢复斜率和绝对值结构
A1	正确函数
B1	完整值域

(b) · 3 marks

$f^{-1}(x) = (4 - 2x)/3$ on $[-1, 2]$.

右支是2−3x/2，一一递减；反解并使用这条支覆盖的输出区间[−1,2]。

M1	选择正确单调支
A1	逆式
B1	逆域

(c) · 4 marks

$x = \pm 4/9$.

外层f的零点是±4/3，但内层f只能输出[−1,2]，所以−4/3不可能。只解f(x)=4/3，得|x|=4/9。

M1	找外层的两候选零输入
B1	以内层值域排除−4/3
M1	解剩下的绝对值式
A1	两根